

EDU 501 Course Outline

EDU 501 — Education Sector Modelling

Course Outline

School	05 Sector Planning — Education
Level	500 (practitioner / advanced)
Hours	55 (indicative band 40–55)
Shape	5 modules · 25 sessions · 8 laboratories · 4 assignments · 1 final assessment
Delivery	Instructor-led — online, blended or in person
Prerequisites	EDIS 101 required. DAT 703 Household Survey Analysis strongly recommended. Regression at the level of a logistic model, and Julia at the level of reading and modifying a script
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-27 · 12-month cycle

What this course is

EDU 501 teaches an education-sector model estate and the judgement required to say what each of its models does and does not establish. The estate is 90 Julia modules across 30 layers: a 72-card specification in 8 families, running from participation and household cost through inclusion, learning and earnings to spatial planning, projection, costing and policy microsimulation; a system engine; nine optimisation programmes; a geographic layer; and six integration bridges.

Six of the eight families are estimated here on real microdata; one is not. No pupil-level assessment data exists on any platform this course reaches, so ten learning models are specified and not estimated. That is the first thing the course teaches.

The rule the course is built on. Every number has a model behind it and a denominator under it. Where the data cannot support the model, the course says so — and teaches the specification anyway.

Who it is for

Ministry of Education planners, statisticians and monitoring officers; education economists; Ministry of Finance sector analysts; national statistical office survey analysts; and modellers joining an education-sector team.

Data and access

No learner touches restricted microdata: no laboratory opens a microdata file, and no identifier and no survey cluster coordinate appears anywhere. Two labels carry through every figure — estimated for a quantity computed from microdata, with its design-based interval and its denominator, and specified for a model the catalogue defines but the data cannot support — and presenting the second as though it were the first is an automatic fail. The spatial layers are public infrastructure, mapped freely.

Structure

Module	Title	Sessions	Laboratory	Hours
M1	Foundations: the estate, the specification, the data, estimation	01–04	01–02	10.8
M2	The catalogue — six families estimated, one specified	05–12	03–05, 07	21.5
M3	Geography, the register and capacity	13–15	06	6.6
M4	Projection, costing and optimisation	16–20	08	9.6
M5	Integration, reconciliation and governance	21–25	08	6.5
	Total	25	8	55.0

The 25 sessions and 8 laboratories, in teaching order, and the subject matter of each:

M1 Foundations: the estate, the specification, the data, estimation — Sessions 01–04 · 10.8 h · Laboratory 01–02

- **01 What this system models** — model-estate architecture; the six layers and what each produces; what each layer can and cannot tell you; the boundary between a specification and its implementation; what has been verified, and by whom
- **02 The specification** — seventy-two models in eight families; what is on a model card; the field that decides estimability; the estimator families the catalogue uses; how the families feed each other
- **03 The data the estate runs on** — household surveys, and what the education recode actually says; the school register; the geographic layers; the canonical schema and how a survey reaches it; what harmonisation cannot fix; what a learner may and may not hold
- **04 Estimation under a complex design** — weights, strata and clusters; the ratio estimator and its linearised variance; singleton strata; the design effect and what it costs; regression under the design; the three scales a coefficient lives on; disclosure

- *Lab 01 The estate and the catalogue* — mapping the estate by layer, reading the catalogue, and triaging every card into estimated, register-derived or specified
- *Lab 02 Design-based estimation* — the design-effect identity, what a cluster correction costs a regression, and the disclosure decision

M2 The catalogue — six families estimated, one specified — Sessions 05–12 · 21.5 h · Laboratory 03–05, 07

- **05 A. Participation, progression and completion** — what the family estimates, and its trap; attendance, enrolment and completion by universe and denominator; net against gross ratios; over-age and repetition; grade flow
- **06 Who is excluded, and where** — out-of-school profiles; the participation gradient by wealth, region, sex and residence; what a recode's own code list will and will not support
- **07 B. Household demand and cost** — household education spending; the denominator that decides the story; the two-part model; an expenditure elasticity below one against a rising budget share
- **08 Catastrophic spending and impoverishment** — the catastrophic-expenditure rate and its threshold; the impoverishing effect; poverty lines and deflation; why the indicator misses the excluded
- **09 C. Inclusion, gender and inequality** — child labour and social inclusion; the concentration index; the Oaxaca-Blinder decomposition and whether the identity closes; inequality of attainment
- **10 D. Learning, the chapter we cannot estimate** — what the ten learning models specify; why an absent dependent variable is not a model that failed; what would change it; writing the data request instead of substituting a proxy
- **11 E. Returns to schooling and selection** — the Mincer equation; the wage sample and who is in it; the two-step selection correction, and the exclusion restriction that identifies it
- **12 What schooling actually changes** — employment, paid employment and formality as distinct outcomes; association against causation; what an employment-rate target would conclude
- *Lab 03 Participation and progression* — what the variable means, net against gross and the gap between them, and the gradient read from both ends
- *Lab 04 Cost, affordability and poverty* — the denominator, the two-part model, and burden and poverty on one dataset
- *Lab 05 Inequality and the gender gap* — the decomposition and whether it closes, inequality of attainment, and the signs
- *Lab 07 Returns, selection and the labour market* — the return and the population it describes, the correction run twice, and three labour outcomes

M3 Geography, the register and capacity — Sessions 13–15 · 6.6 h · Laboratory 06

- **13 F. Geography, where the schools are** — the spatial planning and optimisation family; the school register as a data source; counts by level and ownership; the quarter of schools that record no pupils
- **14 Crowding, capacity and the norm** — four accessibility measures and the four questions they answer; the coverage curve on two denominators; a catchment as a competition; the capacity gap as a function of the norm; straight-line against routed distance
- **15 The spatial data infrastructure** — what the infrastructure holds, and the registry idea; the education GIS database; the coordinate problem stated precisely; licences that travel with the data
- *Lab 06 Geography and accessibility* — coverage computed two ways, the capacity gap at four norms, and what an incomplete geocode costs a map

M4 Projection, costing and optimisation — Sessions 16–20 · 9.6 h · Laboratory 08

- **16 G. Projection, population to requirement** — the population projection; from population to the demand for schooling; what the survey layer contributes here; grade flow and the pupil-years a system will actually carry
- **17 Supply and the binding constraint** — three capacities and a minimum; what the register says about the binding constraint; teachers as a stock, not a target
- **18 Teachers, costing and the envelope** — the costing identity, and where each parameter comes from; the committed run’s financing profile; the national budget as data; a ceiling is not a forecast
- **19 The optimisation suite** — the nine mathematical programmes; the four that carry the capital decision; why the integer ones are harder; infeasibility as information
- **20 Siting, the objective is the decision** — stating an objective before its recommendation; what a siting run needs before it can run; the capacity gap as a capital programme; teacher deployment
- *Lab 08 Projection, optimisation and the handover* — tracing the projection from base-year population to requirement, and showing that changing the objective changes the plan

M5 Integration, reconciliation and governance — Sessions 21–25 · 6.5 h · Laboratory 08

- **21 H. Microsimulation and incidence** — policy microsimulation and its four honest labels; the conservation check; incidence and the accounting unit; the burden, measured
- **22 The CD-CGE handover** — the design rule, and the exchange contract; what the committed run actually supplies; the two fields that matter most and are empty; what has to be in the handover note
- **23 Poverty, the MTEF and cross-sector** — the fiscal envelope and the approved budget as data; household spending as part of the envelope; the cross-sector schema and the gate that refuses a causal claim; a null result reported as null
- **24 Four sources, four numbers** — the four kinds of disagreement — definitional, temporal, coverage and denominator; what not to do; the three-sentence reconciliation note
- **25 Governance, disclosure and the handover** — what may leave the platform; the disclosure threshold; licences that travel with the data; reproducibility defined, and what it is not; the interpretation rules that are automatic fails; the four sentences
- *Lab 08 The handover, concluded* — the exchange contract read against its payload, and the four reconciliation pairs

Laboratory 08 is a single sitting run across M4 and M5: its first two parts close the planning chain, and its third opens the handover. Five of the course’s sixteen registered limitations were found by re-deriving a shipped figure rather than by reading documentation — an attendance code that appears in neither survey; a register that locates half the schools it counts; a selection correction that triples an estimate and is not identified; four sources that do not reconcile; and a published audit figure carrying the wrong denominator. Each is taught where it arises.

What a graduate can do

1. Locate any model in the estate and say which of the six layers produced any given number.
2. Distinguish a model that failed from one that cannot be estimated, and write the data request implied.
3. Read a survey recode’s own code list before using a variable, and report a figure with the label the data support.

4. Produce a design-based estimate with its interval, denominator, design effect, effective sample size and disclosure decision.
5. Distinguish attendance, enrolment and completion by universe and denominator, and explain a gross ratio above one.
6. Compute household education spending per student and per household, and recognise a denominator error in someone else's published figure.
7. Compute a catastrophic-expenditure rate and an impoverishing effect, decompose a group gap, and state what the unexplained term is not.
8. Quote a return to schooling with the population it describes, and recognise a selection correction that is impressive because it is unidentified.
9. Quantify what an incomplete geocode costs a map, compute a capacity gap with its norm, and state an optimisation objective before its recommendation.
10. Read a macro-exchange payload against its contract and write the handover note — every empty field named, each cross-source disagreement classified, and what the analysis does not establish.

The twenty-three formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	15 %	125 items across 16 modules and six question types
Laboratory exercises	30 %	8 laboratories, equally weighted, marked on model competence, interpretation, honesty about limits and communication
Applied assignments	25 %	The estate note and catalogue triage; one family estimated and reported; a map that says what it can support; the handover note
Final assessment	30 %	Three hours, open book: foundations; estimation and affordability; geography and optimisation; and a handover note written for a named recipient

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance sidecar is in the output: an irreproducible result is no result.

Materials and set-up

A 250-page Technical Manual; 25 lecture decks and 8 laboratory decks; a Laboratory Manual, Instructor Guide and Learner Workbook; a 125-item quiz bank with four assignments, a final assessment and a

marking guide; 54 datasets; 9 code exercises; 3 case studies; and the LMS package. The README lists every file.

Before the first laboratory, learners need the estate instantiated at `D:/EDIP/models/EducationModelling`. An Application-Control policy on this host blocks native package images, so every Julia command passes `--pkgimages=no`; the full sequence is in the Laboratory Manual's pre-flight section.

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